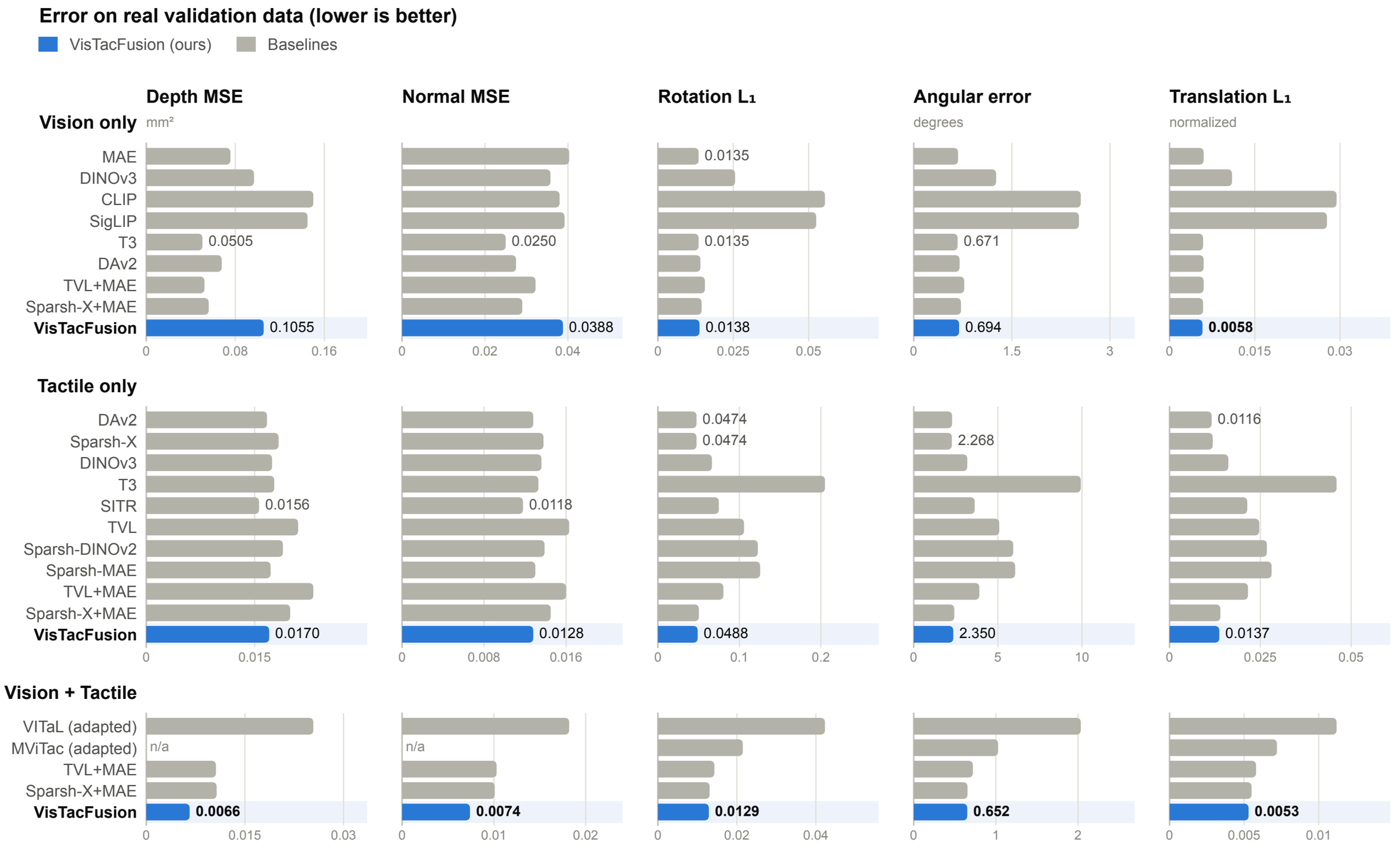
Error on real validation data (lower is better)
VisTacFusion (ours)
Baselines
Depth MSE
mm²
Normal MSE
Rotation L₁
Angular error
degrees
Translation L₁
normalized
Vision only
MAE
DINOv3
CLIP
SigLIP
T3
DAv2
TVL+MAE
Sparsh-X+MAE
VisTacFusion
0.0505
0.1055
0.0250
0.0388
0.0135
0.0135
0.0138
0.671
0.694
0.0058
0 0.08 0.16
0 0.02 0.04
0 0.025 0.05
0 1.5 3
0 0.015 0.03
Tactile only
DAv2
Sparsh-X
DINOv3
T3
SITR
TVL
Sparsh-DINOv2
Sparsh-MAE
TVL+MAE
Sparsh-X+MAE
VisTacFusion
0.0156
0.0170
0.0118
0.0128
0.0474
0.0474
0.0488
2.268
2.350
0.0116
0.0137
0 0.015
0 0.008 0.016
0 0.1 0.2
0 5 10
0 0.025 0.05
Vision + Tactile
VITaL (adapted)
MViTac (adapted)
TVL+MAE
Sparsh-X+MAE
VisTacFusion
n/a
n/a
0.0066
0.0074
0.0129
0.652
0.0053
0 0.015 0.03
0 0.01 0.02
0 0.02 0.04
0 1 2
0 0.005 0.01